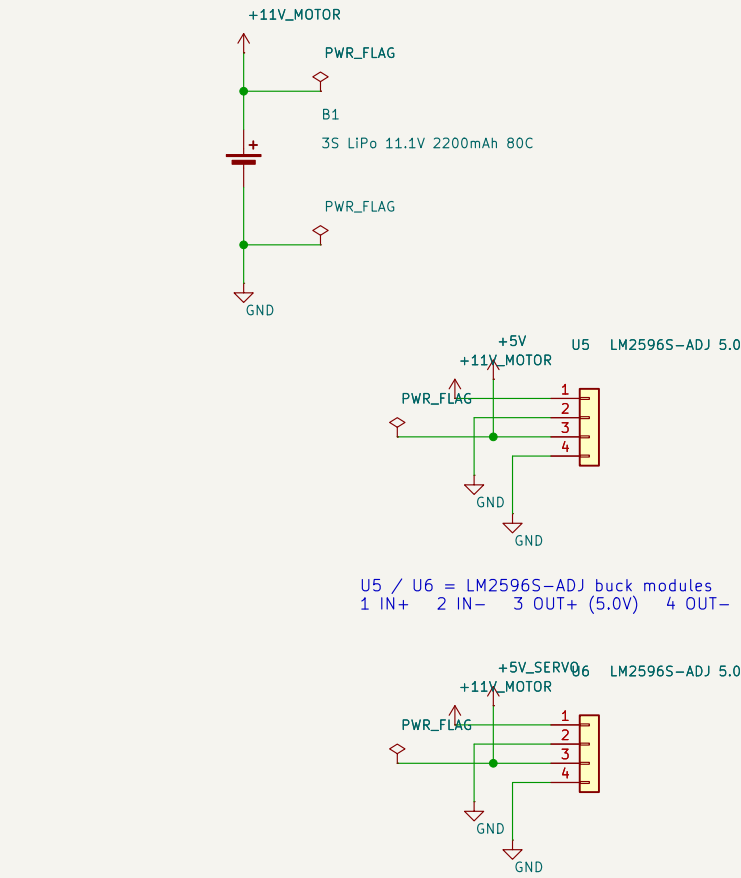


SINGULARITY ROBO – MAIN ELECTRICAL SCHEMATIC

1 . POWER / BATTERY / REGULATORS



POWER ARCHITECTURE – SINGULARITY ROBO
3S LiPo 11.1V (12.6V max) 2200mAh 80C
A) +11V_MOTOR -> U2 L298N VS (motors)
B) +11V_MOTOR -> U5 LM2596S-ADJ -> +5V
C) +11V_MOTOR -> U6 LM2596S-ADJ -> +5V_SERVO
(dedicated high-current servo rail)

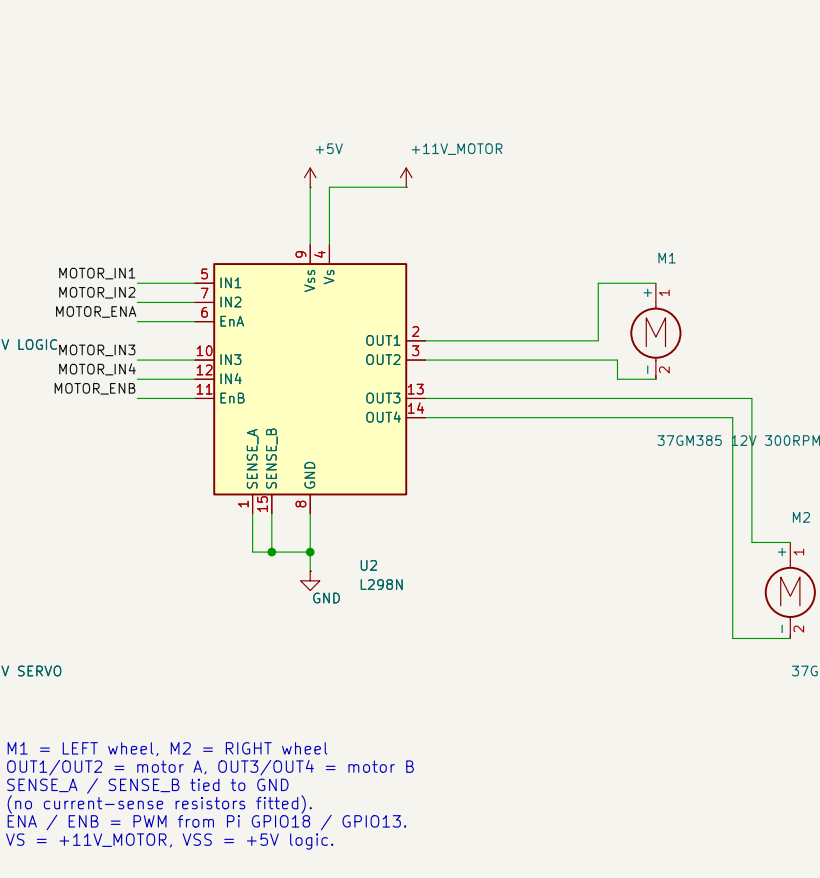
+5V feeds: Raspberry Pi, PCA9685 VDD,
HC-SR04, ESP32-CAM, TFT, FlySky RX,
Bluetooth speaker power.

+5V_SERVO feeds: servo V+ rail J4-J7 ONLY.

*** DO NOT CONNECT 11.1V TO RASPBERRY PI ***
U1 5V pins are fed ONLY from U5 +5V.

ALL GROUNDS COMMON: battery - = GND.
Motor 11.1V / logic 5V / servo 5V rails
kept separate; PWR_FLAG marks each source.

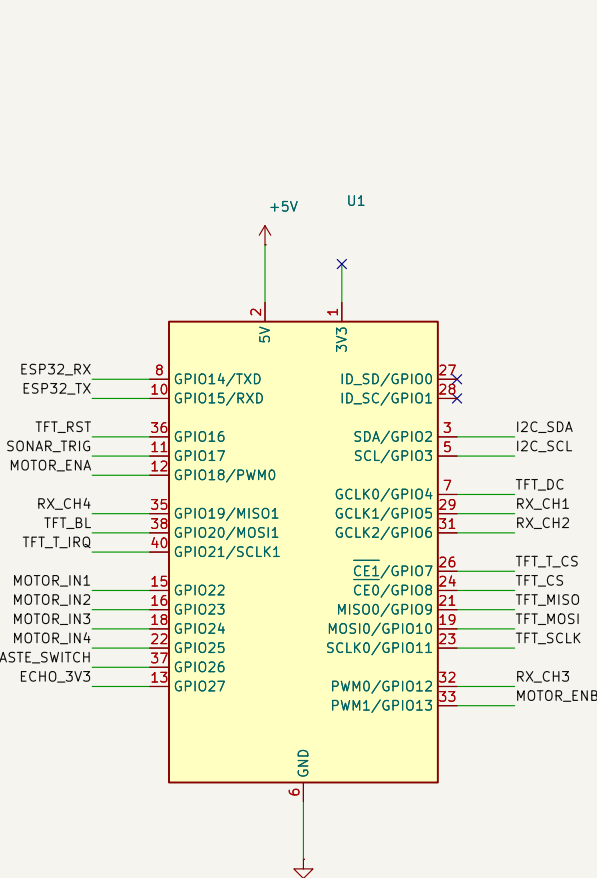
2 . MOTOR DRIVER L298N



M1 = LEFT wheel, M2 = RIGHT wheel
OUT1/OUT2 = motor A, OUT3/OUT4 = motor B
SENSE_A / SENSE_B tied to GND
(no current-sense resistors fitted).
ENA / ENB = PWM from Pi GPIO18 / GPIO13.
VS = +11V_MOTOR, VSS = +5V logic.

U6 = DEDICATED SERVO BUCK, high current.
Servo rail never loaded by Raspberry Pi.

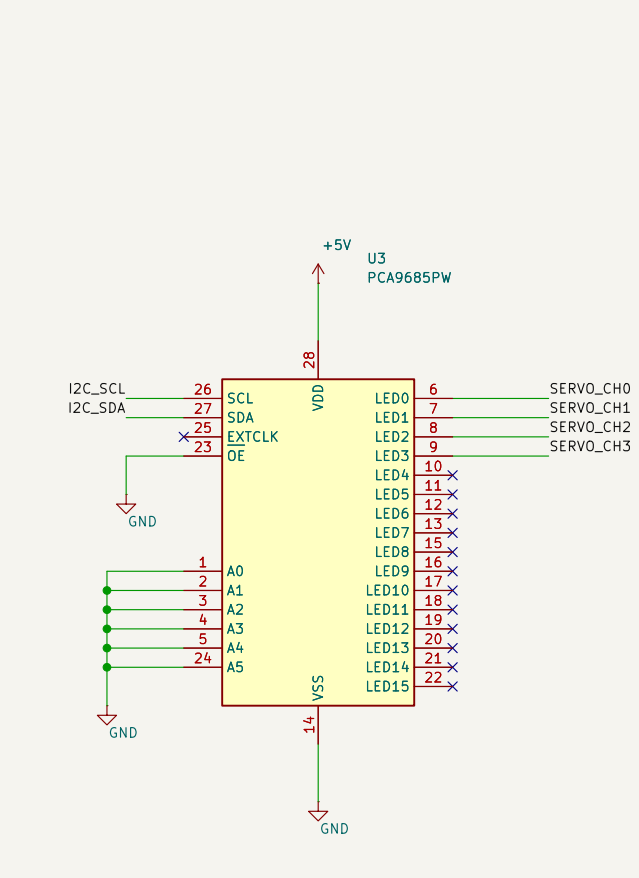
3 . RASPBERRY PI 3 MODEL B+ (U1)



Raspberry Pi 3 Model B+

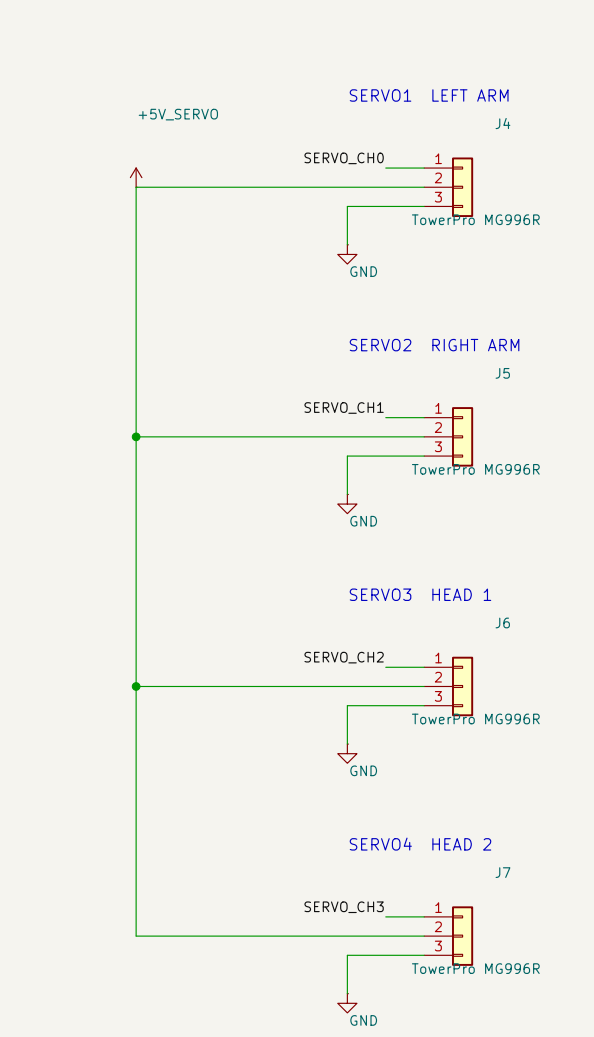
U1 GPIO MAP (all 26 GPIOs used)
I2C GPIO2/SDA + GPIO3/SCL -> U3 PCA9685
SPI0 GPIO10/9/11 + CE0/CE1 -> J2 TFT
CE0 = TFT_CS, CE1 = touch T_CS
ENA/ENB = PWM GPIO18 / GPIO13
UART GPIO14/15 <-> U4 ESP32-CAM (crossed)
5V from U5 ONLY – NEVER 11.1V
BT audio -> Dubstep Pop 600 (wireless)

4 . PCA9685 16-CH I2C PWM (U3)



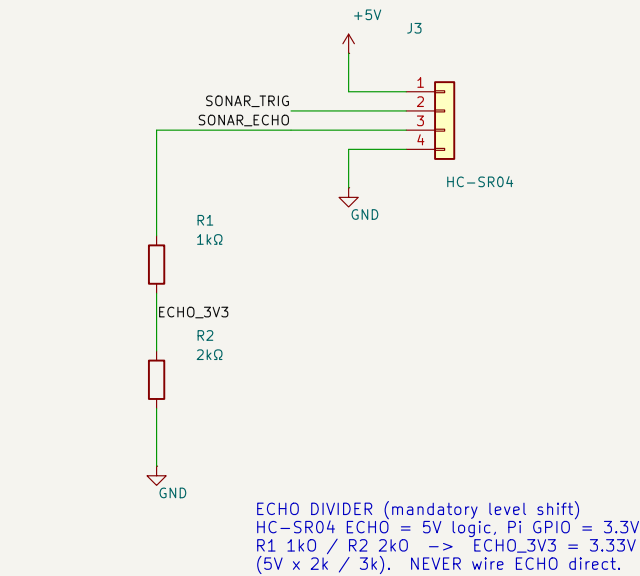
U3 PCA9685PW: A0-A5 = GND (addr 0x40).
-OE = GND (outputs enabled), EXTCLK = NC.
VDD = +5V logic. Servo V+ rail = +5V_SERVO
routed to J4-J7 headers. LED0-3 = CH0-3.

5 . MG996R SERVO INTERFACES



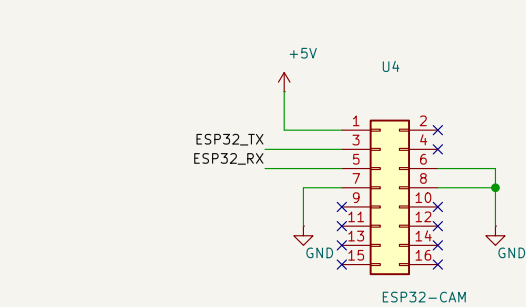
SERVO POWER RAIL: +5V_SERVO from U6 only.
Header: 1 SIGNAL . 2 +5V_SERVO . 3 GND
(common ground). NEVER from Pi 5V GPIO.

6 . HC-SR04 ULTRASONIC + ECHO DIVIDER



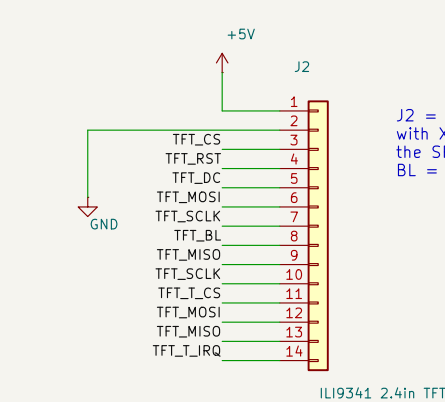
ECHO DIVIDER (mandatory level shift)
HC-SR04 ECHO = 5V logic, Pi GPIO = 3.3V
R1 1kΩ / R2 2kΩ -> ECHO_3V3 = 3.33V
(5V x 2k / 3k). NEVER wire ECHO direct.

7 . ESP32-CAM VISION MODULE



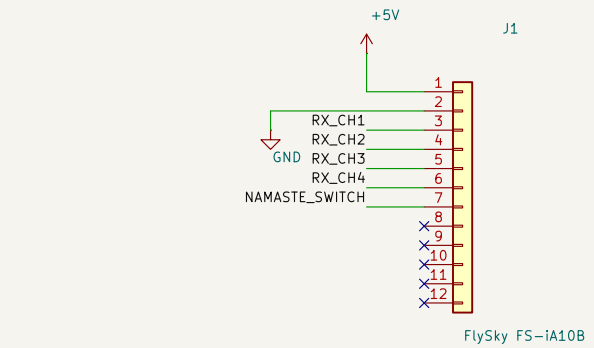
U0T (GPIO1) = TX -> Pi RXD = ESP32_TX
U0R (GPIO3) = RX <- Pi TXD = ESP32_RX
Camera / person detection / inference on Pi
5V from +5V rail, GND common, rest NC.

8 . ILI9341 2.4in SPI TFT + TOUCH



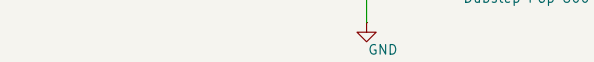
J2 = 2.4in ILI9341 240x320 SPI display
with XPT2046 resistive touch. Touch shares
the SPI bus. T_CS on CE1 (GPIO7).
BL = GPIO20 backlight, VCC = +5V.

9 . FLYSKY RC RECEIVER



FLYSKY FS-i6X – WIRELESS TRANSMITTER
(2.4 GHz) binds to J1 FS-IA10B receiver.
CH1-CH4 drive / camera, CH5 = NAMASTE.
Pi reads the PWM channel pulses on GPIO.

10 . BLUETOOTH AUDIO



Dubstep Pop 600 speaker: audio link is
BLUETOOTH (wireless) from the Raspberry Pi.
Only +5V / GND are wired (power/charge).